

A Hierarchical Retail World Model for Interpretable Demand Decomposition

An operations-research working article prepared in IMRAD style. All numerical examples are fictional but calibrated to plausible retail magnitudes.

Abstract

Retail demand analytics often starts from dashboards or predictive algorithms. This article proposes the reverse order: first define the retail world, then specify its hierarchical mechanisms and event chains, then derive metrics and finally assign classical, interpretable algorithms. The proposed model represents retail as a time-indexed system of product flow, store availability, commercial mechanics and customer response. Observed sales changes are treated as downstream effects that can be decomposed into price, quantity, distribution, availability, promotion, category demand, competitor and residual components. A fictional New York retail case demonstrates that a sales increase may be driven primarily by distribution and shelf availability, while price discounting compresses revenue quality. The contribution is a non-LLM, non-black-box analytical architecture suitable for strategic monitoring and diagnostic decomposition rather than dashboarding or optimization.

Keywords: retail analytics; operations research; demand decomposition; causal graph; metric ontology; promotion analysis; stock availability; interpretable models.

1. Introduction

A common failure mode in retail analytics is to move directly from raw transaction data to dashboards, forecast models or optimization routines. This prematurely hides the data-generating process. For strategic monitoring, the more important problem is diagnostic: when sales in a market such as New York increase, the analyst needs to know whether the increase was caused by baseline demand, store availability, distribution expansion, promotion, price, portfolio mix, competitor movements, or unexplained noise.

This article treats retail as a structured operations system rather than as a flat table of KPIs. The starting point is an entity ontology: markets, stores, SKUs, shelves, distribution centers, stock states, promotions, prices, customers and transactions. The next layer is a hierarchy of mechanisms: logistics creates product flow into stores; store stock and shelf execution create availability; price and promotion modify the customer's economic signal; customers respond through traffic, conversion, substitution and basket quantity; only then do revenue, units, margin and share become observable. This order is consistent with the use of explicit causal assumptions in graphical models [1] and with operational evidence that out-of-stock conditions materially affect retail outcomes [2].

The purpose of the proposed system is not to recommend actions, optimize the portfolio, or render a BI interface. It is to construct a semantic and causal measurement system that can answer structured diagnostic questions. The system is deliberately restricted to classical and interpretable methods: deterministic decomposition, robust rules, linear or generalized linear models, fixed effects, time-series decomposition, changepoint detection and quasi-experimental comparisons. No LLM is used for analysis, and no black-box model is required.

Research question.

How can a retail analytics system be organized so that observed changes in sales, units or share can be decomposed into interpretable upstream drivers before any algorithmic detection layer is selected?

2. Methods

2.1 Entity ontology

The ontology defines what exists in the retail world and at which granularity it is observed. The recommended analytical grain is SKU-store-time, usually daily or weekly for operations and monthly for executive monitoring. Higher-level views are aggregations over this grain, not separate realities.

Layer	Entities	State variables	Primary questions
Market and context	country, region, city, catchment, competitor set	category demand, local macro index, weather, competitor price and promo	Is the market growing or contracting independently of our execution?
Network	DC, store, channel, shelf location	lead time, receipts, on-hand inventory, shelf capacity, planogram	Did product flow and availability precede the observed outcome?
Product	category, brand, product family, SKU, pack size	assortment status, lifecycle, substitution group, margin class	Which SKU or family explains the aggregate movement?
Commercial mechanics	base price, net price, promotion, display, media, coupon	price index, depth of discount, promo flag, exposure intensity	Was growth structural or mechanically induced?
Customer and basket	customer segment, visit, basket, transaction	traffic, conversion, units per basket, switching, stock-up	Did demand rise, shift, or get pulled forward?
Outcome	transaction line, SKU-store-period aggregate	units, revenue, margin, share, residual	What changed, and what upstream event explains it?

Table 1. Retail entity ontology. The outcome layer is deliberately last: sales are an effect, not the first object of explanation.

2.2 Hierarchical mechanism model

The model is hierarchical because retail mechanisms operate at different levels and lags. A market shock may affect all stores in a city. A logistics failure may affect a subset of stores. A promotion may affect only participating SKUs and stores. A customer substitution response may occur inside a product family even when category demand is unchanged.

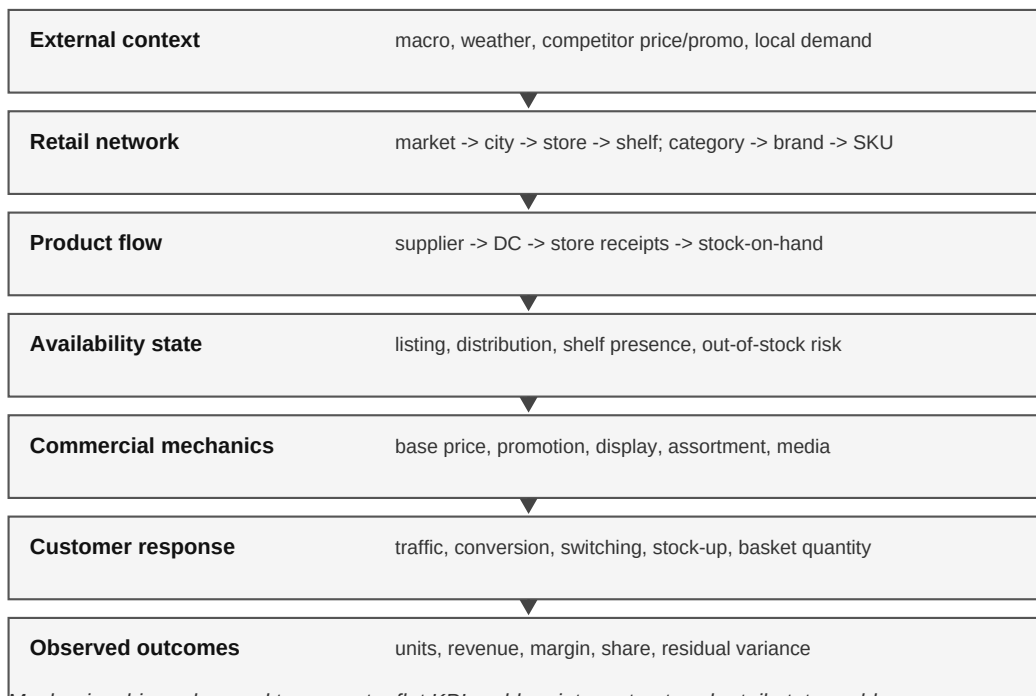
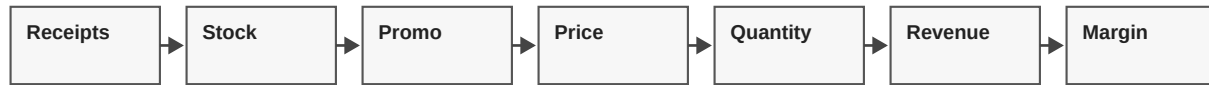


Figure 1. Mechanism hierarchy used to convert a flat KPI problem into a structured retail-state problem.

2.3 Event-chain model

The system should store not only metric values but also the approximate order of events. This is essential because the same sales pattern has different interpretations depending on whether stock increased before promotion, price changed before quantity, or competitor actions preceded share loss.



The model does not assume that revenue moves first. It reconstructs the path from upstream states to outcomes.

Example: higher receipts -> higher shelf availability -> promo price cut -> higher units -> revenue may rise, stay flat, or fall.

Figure 2. Example event chain. The same observed revenue change can be caused by different upstream mechanisms.

Event family	Typical order	Expected lag	Diagnostic implication
Replenishment and logistics	order -> shipment -> store receipt -> backroom stock	days to weeks	If sales fall after receipts and availability fall, the primary hypothesis is supply/availability, not demand.
Shelf and availability	stock-on-hand -> shelf presence -> customer sees item	same day to days	Availability is a gate: demand cannot be observed when the item is absent.
Commercial mechanics	promotion planned -> price/display changes -> exposure	same day to two weeks	A price cut can raise units while reducing average revenue per unit and margin.
Customer response	traffic -> conversion -> basket quantity -> switching	same day	Volume can rise through new demand, switching, or stock-up; these are not equivalent.
Competitive context	competitor promo/price -> relative value changes -> share response	days to weeks	Share loss with stable internal execution suggests external pressure.

Table 2. Approximate event order. The order is a modeling assumption and should be made explicit for each category and market.

2.4 Metric production model

Metrics are not independent dashboard tiles; each metric observes a part of the retail mechanism. Revenue is a downstream product of price and quantity. Quantity is constrained by distribution and availability before being affected by price, promotion, customer response and competitors. Let R be net revenue, P average net price per unit and Q units sold for store s, SKU k and period t:

$$\log R_{\text{skt}} = \log P_{\text{skt}} + \log Q_{\text{skt}}$$

Quantity is then decomposed through upstream multiplicative factors:

$$\log Q_{\text{skt}} = \log B_{\text{skt}} + \log G_{\text{skt}} + \log A_{\text{skt}} + \log M_{\text{skt}} + \log C_{\text{skt}} + e_{\text{skt}}$$

where B is baseline or category demand, G is distribution/listing coverage, A is availability, M is commercial mechanics such as promotion and display, C is competitive response and e is residual variation. This is not a claim of automatic causality. It is a structural bookkeeping model that forces the analyst to state assumptions before applying algorithms.

Observed metric	Mechanism observed	Primary decomposition role
Revenue	price multiplied by units	separates value growth from volume growth
Units	customer demand after availability and mechanics	main volume response metric
Average net price	price and discounting	detects price compression and promo mechanics
Distribution percent	stores where SKU is listed or sold	detects expanded or lost reach
Availability / OOS proxy	whether demand can be observed	detects supply and shelf execution constraints
Promo exposure	discount/display/media mechanics	separates tactical lift from structural demand
Category demand	market-level baseline	controls for market growth or decline
Relative price index	our price versus competitor price	detects external price pressure

Table 3. Metric production model. Each metric is attached to a mechanism in the retail hierarchy.

2.5 Decomposition procedure

For an observed movement between period 0 and period 1, the system performs a sequential counterfactual decomposition in the order implied by the event model. In log form, revenue movement is decomposed as:

$$\Delta \log R = \Delta \log P + \Delta \log B + \Delta \log G + \Delta \log A + \Delta \log M + \Delta \log C + \Delta e$$

The order is not purely statistical; it is an operations assumption. For example, stock availability should usually precede customer response. To assess sensitivity, the system can recompute the bridge under a small set of admissible alternative orders and report whether the conclusion is stable. This preserves interpretability while acknowledging that retail processes are not perfectly observed.

2.6 Scenario layer before algorithms

Algorithms are selected only after the system defines canonical scenarios of metric behavior. This avoids fitting models to metrics before specifying what those metrics mean.

Scenario	Metric pattern	Business meaning	Candidate classical methods
Healthy baseline	metrics remain inside expected seasonal bands	normal state; no diagnostic alert	STL/ETS baseline; robust control bands
Availability-driven change	stock or distribution changes before units	supply or execution explains sales movement	event-chain decomposition; fixed effects regression
Promo-only growth	units lift during promo but baseline does not improve	tactical volume without structural demand	interrupted time series; distributed-lag regression
Price compression	price down; quantity up; revenue or margin weak	volume bought with discounting	log revenue bridge; elasticity regression
Competitor pressure	relative price worsens; share falls	external value proposition risk	controlled regression; DiD if matched markets exist
Cannibalization	one SKU rises while adjacent SKUs fall and category is flat	internal substitution, not net growth	portfolio flow matrix; SKU-family fixed effects
Structural break	level or trend changes persist after event	new regime; old thresholds obsolete	PELT changepoint; Chow test; rolling coefficients
Market divergence	same SKU behaves differently by market	local execution or competition	hierarchical reconciliation; market clustering

Table 4. Scenario layer. This layer sits between metric ontology and algorithm selection.

2.7 Algorithmic setup

The final analytical layer uses only interpretable classical algorithms. Seasonal-trend decomposition separates baseline movement from seasonal variation; STL is a standard approach for decomposing a series into trend, seasonal and residual components [5]. ETS and ARIMA-family models are used for transparent baselines and forecast residuals [6]. PELT detects structural changes in level or variance with efficient changepoint search [8]. Fixed effects, distributed-lag and difference-in-differences regressions estimate interpretable effects when comparison stores, SKUs or markets exist. Bayesian structural time-series methods can be used for intervention impact when synthetic controls are available, though this article treats them as optional because they require stronger modeling assumptions [9]. Hierarchical forecast reconciliation is relevant when city, store, category and SKU estimates must remain additive across levels [7].

Problem	Recommended method	Reason for inclusion
Seasonal baseline	STL + ETS/SARIMAX	separates trend, seasonality and residual before diagnostic interpretation
Event impact	OLS/GLM with fixed effects; distributed lags	coefficient-level interpretation by price, promo, stock and store
Promo lift	pre/post with controls; DiD where possible	distinguishes promotional mechanics from general market movement

Problem	Recommended method	Reason for inclusion
Structural breaks	PELT, Chow test, rolling coefficients	detects persistent regime changes
Outliers	median absolute deviation, Hampel filter, robust z-score	transparent anomaly rules rather than opaque anomaly models
Hierarchy coherence	bottom-up aggregation; least-squares reconciliation	ensures SKU, store, city and market estimates sum consistently

Table 5. Interpretable algorithm layer. The algorithms serve the scenario model; they do not define it.

3. Results

3.1 Fictional New York diagnostic case

A fictional case was constructed for an energy-drink SKU family sold in New York grocery and convenience stores. The business question is: Why did sales in New York increase in April compared with March? The example deliberately includes an upstream operational change before the promotional event.

Variable	March	April	Change
Stores with listing	84 of 100	91 of 100	+8.3 percent relative
Shelf availability	90 percent	96 percent	+6.7 percent relative
Promo participation	8 percent of stores	45 percent of stores	expanded promo exposure
Average net price	USD 3.49	USD 3.29	-5.7 percent
Category traffic index	1.00	1.02	+2.0 percent
Units sold	205,000	257,000	+25.4 percent
Net revenue	USD 715,450	USD 845,530	+18.2 percent

Table 6. Fictional New York case. Units rise faster than revenue because average price falls during promotion.

3.2 Decomposition of the sales increase

The revenue ratio is $845,530 / 715,450 = 1.182$, or +18.2 percent. In log terms the movement is 16.7 log points. The decomposition attributes the movement as follows:

Driver	Ratio	Log-point contribution	Interpretation
Category traffic	1.020	+2.0	moderate market tailwind
Distribution expansion	1.083	+8.0	more stores carried the SKU
Availability recovery	1.067	+6.5	fewer lost sales from stock/shelf absence
Promo response	1.060	+5.8	moderate unit lift from promo mechanics
Competitive effect	1.000	+0.0	no material competitor movement assumed
Average price	0.943	-5.9	discount compresses revenue per unit
Residual	1.003	+0.3	unexplained but small
Total	1.182	+16.7	equals +18.2 percent revenue growth after exponentiation

Table 7. Log revenue bridge. The main positive drivers are distribution and availability; price is a negative value driver.

The diagnostic statement produced by the system is therefore not simply 'sales increased'. A more accurate statement is: New York revenue increased because the SKU reached more stores and was more available on shelf, with additional promotional unit lift; however, the price discount reduced revenue quality, so units increased more than sales value. This is a mixed scenario: availability-driven growth plus promo-assisted growth plus price compression.

3.3 Scenario classification

The case activates three scenarios. First, availability-driven change is high confidence because store listing and shelf availability improved before the observed unit increase. Second, promo-assisted growth is moderate confidence because a broad promotion coincided with incremental unit response; without matched non-promoted stores, this should not be treated as a fully causal estimate. Third, price compression is high confidence because average net price dropped while units increased. The case does not support a pure demand breakthrough because a large part of the lift can be explained by upstream operational and commercial mechanics.

4. Discussion

The central implication is that retail analytics should model the production of metrics before selecting algorithms. A dashboard may show that New York sales rose, but a mechanism model asks what had to happen upstream for that observation to appear. This is especially important in retail because availability gates demand observation: when an item is not on shelf, lost demand is not directly recorded. Similarly, a promotion can create apparent growth while reducing price per unit, margin or future demand through pull-forward effects. Prior marketing research shows that price promotions can affect category demand, brand choice and purchase quantity in non-equivalent ways [3,4].

The proposed architecture is intentionally conservative. It prefers weaker but interpretable claims over stronger but opaque predictions. A fixed-effects regression with explicit controls may underfit compared with a black-box learner, but it preserves the ability to say whether a unit change was associated with price, distribution, availability or local market conditions. The decomposition bridge is also auditable: every contribution can be traced to an observable or explicitly assumed factor.

There are limitations. First, causal interpretation depends on the validity of the event order and adjustment assumptions. If promotion allocation is correlated with unobserved store-level demand, naive regression will overstate promo impact. Second, availability proxies can be noisy: point-of-sale data observes sold units, not necessarily unmet demand. Third, decomposition is not unique when drivers interact. Price, promo and availability can reinforce one another. The system should therefore report confidence classes and sensitivity checks rather than a single overconfident explanation.

For an operations-research implementation, the practical object is a structured diagnostic engine. It is not an action layer. It does not decide what to do. It reconstructs likely mechanisms behind observed metric changes and provides a disciplined basis for later decision-making, experimentation or optimization.

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Further work

The next research step is to formalize the retail world model as an explicit directed acyclic graph for each category and market. Each node should represent a state variable, each edge should include an expected lag, and each metric should be mapped to the node it observes. This would allow analysts to state which causal paths are assumed, which are testable and which remain unobserved.

Second, the decomposition procedure should be validated on real SKU-store-time data with known interventions: promotion launches, distribution changes, stock recovery events and competitor price shocks. The validation target should not only be forecast accuracy; it should also include diagnostic correctness, such as whether the system identifies a stock-driven decline as availability loss rather than demand erosion.

Third, the scenario catalog should be expanded into a governed taxonomy with confidence rules. For example, 'promo-only growth' should require both a promotion-period lift and weak post-promo carryover, while 'availability-driven growth' should require evidence that stock or distribution improved before quantity increased.

Fourth, the system should support sensitivity analysis over alternative event orders. Retail mechanisms interact, and a single decomposition order can overstate one driver when price, promo and availability move together. Reporting a range over admissible orders would make the model more credible.

Finally, only after the diagnostic layer is validated should an action or optimization layer be considered. The present article intentionally stops before recommendations: its role is to make the data-generating process legible, not to prescribe decisions.